

Sai Sri Harsha Guddati

saisriharshaguddati1@gmail.com — (352) 665-8709 — Gainesville, FL — [linkedin.com/in/sai-sri-harsha-guddati](https://www.linkedin.com/in/sai-sri-harsha-guddati) — github.com/saisriharsha19 — <https://saisriharsha.me>

EDUCATION

University of Florida

Master of Science in Computer Science, GPA: 3.9/4.0

Coursework: Distributed Systems, Advanced Data Structures & Algorithms, Database Systems

Jan. 2024 – Dec. 2025

Gainesville, FL

WORK EXPERIENCE

University of Florida, Information Technology

Software Engineer Intern, Platform and Distributed Systems

May. 2025 - Dec. 2025

Gainesville, FL

- Built and operated backend services for **Navigator Chat** and **Prompt Builder** using FastAPI and Celery, handling asynchronous AI inference for **30,000+ active users**.
- Migrated **Navigator Assistant** services from bare-metal to Kubernetes; authored Helm charts and GitHub Actions pipelines that reduced deployment time from **2 hours to 45 minutes** and improved rollout reliability.
- Reduced API P95 latency from **800ms to 480ms** by instrumenting custom Prometheus metrics and diagnosing PostgreSQL connection pooling contention via Grafana dashboards.
- Designed Redis-based caching for embedding and inference results in the RAG pipeline, eliminating redundant LLM calls and reducing external API spend by **40%**.

Tata Consultancy Services

Software Development Engineer

Aug. 2022 – Dec. 2023

Hyderabad, India

- Developed a semantic search backend using **Flask** and **Pinecone** on Azure, enabling natural language queries across **10M+ enterprise documents**.
- Owned CI/CD pipelines in GitLab for multiple backend services across Dev, QA, and Prod; enforced approval gates and automated tests, resulting in **zero Sev-1 production incidents** over 12 months.
- Built Pytest-based test suites with mocked external dependencies, increasing code coverage to **85%** and preventing regressions during frequent releases.
- Optimized PostgreSQL performance by introducing composite indexes on high-cardinality columns, reducing average query latency by **50%** (1.2s to 0.6s).

Internshala

Software Development Engineer Intern

May 2021 – Aug. 2021

Hyderabad, India

- Built Java Spring Boot services for payment processing; replaced blocking I/O with asynchronous execution, improving throughput and maintaining stable latency under peak traffic.
- Implemented Apache Kafka consumers to decouple notification workflows from the transaction ledger, ensuring reliable message delivery during traffic spikes.

PROJECTS

Distributed Cache System with Consistent Hashing

Python — Redis — Docker — Flask

- Designed a distributed caching layer across a **5-node Redis cluster** using consistent hashing to enable horizontal scaling with minimal key redistribution.
 - Implemented failure detection and automatic rerouting using a gossip-style health check mechanism, sustaining **99.95% availability** during node failure simulations.
- ### Delphi Pascal Interpreter

Java — ANTLR — Compiler Design

- Built a lexer and parser for a subset of Delphi/Pascal using **ANTLR**, implementing an AST-based execution engine with lexical scoping and memory management.
- Designed a hash map-backed symbol table enabling **O(1)** identifier lookup, improving execution performance for nested control flow and recursive functions.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript/TypeScript, C++, Bash

Core: Data Structures, Algorithms, Distributed Systems, System Design

Backend: FastAPI, Flask, Spring Boot, REST APIs, Microservices

Databases: PostgreSQL, MongoDB, Redis, MSSQL, Pinecone

Infrastructure: Docker, Kubernetes, AWS, Azure, Terraform, Helm, GitHub Actions

Observability & Testing: Prometheus, Grafana, Pytest, Logging, Monitoring

Messaging: Apache Kafka, RabbitMQ, Celery

AI Systems: LLM APIs, RAG Architectures, PyTorch